

A Data-Oriented Conceptual Model and Hybrid TDNN–BiLSTM Framework for Context-Aware Speaker Verification in Smart Environments

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Abstract— Smart environments and ambient assisted living systems increasingly rely on speaker verification as an unobtrusive mechanism for user authentication. However, requirements for such systems are often captured only in language-based conceptual models, which suffer from ambiguity, inconsistency, and weak linkage to data-intensive AI implementations. This gap makes it difficult to systematically conceptualize, manage, and build big-data-driven, context-aware speaker verification services for future cyber societies. In this paper, we propose a data oriented conceptual model for context-aware speaker verification that addresses key issues of language-based conceptual modeling. As a concrete instantiation of the proposed conceptual model, we designed a hybrid TDNN–BiLSTM speaker embedding framework that integrates Multi-Head Attention (MHA) pooling and an Additive Angular Margin (AAM) softmax (ArcFace) objective. TDNN layers efficiently capture local acoustic patterns, such as formant transitions and phoneme-level cues, whereas the BiLSTM module models long-range sequential dependencies, such as speaking rhythm and prosody. The MHA pooling layer aggregates frame-level features into fixed-dimensional embeddings, while the AAM-softmax objective optimizes angular margins between speakers in the embedding space, enhancing intra-speaker compactness and inter-speaker separation for open-set verification. Experiments on the VoxCeleb1 test set demonstrate that the proposed implementation achieves an Equal Error Rate (EER) of 0.97% and a minimum Detection Cost Function (minDCF) of 0.0717, indicating that the instantiated framework can deliver highly discriminative embeddings within the proposed conceptual design. These results illustrate how a well-defined conceptual model can effectively guide the development of robust, AI-based speaker verification services for sensor-rich smart environments in the physical world and the future cyber society.

Keywords— *Speaker verification, TDNN–BiLSTM, Multi-Head Attention.*

I. INTRODUCTION

Speaker verification (SV) has become a core component of modern biometric systems, with applications in secure voice

authentication, telephone banking, forensic analysis, smart homes, and voice-controlled assistants such as Amazon Alexa, Apple Siri, and Google Assistant. Given two speech utterances, an SV system extracts fixed-dimensional speaker embeddings and measures their similarity to determine whether both belong to the same individual. Classical approaches based on front-end factor analysis and i-vectors [1] provided a strong foundation for text-independent SV, but their performance is limited in highly variable, real-world conditions. With the advent of deep learning, the x-vector framework [2] introduced Time-Delay Neural Networks (TDNNs) with statistics pooling, leading to substantial gains on large-scale datasets such as VoxCeleb [10]. ECAPA-TDNN [3] further extended this paradigm by incorporating Res2Net-based blocks, Squeeze-and-Excitation channel attention, and multi-layer feature aggregation, establishing strong benchmarks for text-independent SV. Variants such as densely connected TDNNs [11] and multi-level pooling architectures [5] have been proposed to enrich temporal modeling while preserving the core x-vector structure. In parallel, self-attentive pooling [7] and self multi-head attention mechanisms [8] have been explored to learn more informative temporal aggregations, and TDNN–LSTM hybrids [6] have been investigated to capture longer-range temporal dependencies. Margin-based objectives such as ArcFace (Additive Angular Margin Softmax) [4] further enhance discriminability by enforcing angular separation between speaker classes.

In this work, we propose a hybrid TDNN–BiLSTM architecture that integrates Multi-Head Attention (MHA) pooling and an Additive Angular Margin (AAM) softmax objective for text-independent SV. The TDNN layers capture local acoustic patterns, while the BiLSTM module models long-range temporal structure; MHA pooling aggregates frame-level representations using multiple attention heads, and AAM-Softmax encourages intra-speaker compactness and inter-speaker separation. We evaluate the proposed framework on the VoxCeleb1 test set [10], demonstrating competitive

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performance in terms of Equal Error Rate (EER) and minimum Detection Cost Function (minDCF).

II. RELATED WORK

Early speaker verification systems relied on generative modeling and factor analysis. The i-vector framework [1] represented each utterance as a low-dimensional vector derived from a GMM-UBM, combined with PLDA back-ends. Although effective, i-vectors struggle with highly variable, in-the-wild speech. Deep learning shifted the focus to discriminatively trained embeddings. The x-vector architecture [2] uses TDNN layers with statistics pooling and a softmax classifier over speakers. Trained on large-scale corpora such as VoxCeleb [10], x-vectors significantly improved robustness for text-independent SV. The ECAPA-TDNN architecture [3] further enhanced this paradigm by incorporating Res2Net-style blocks, Squeeze-and-Excitation channel attention, and multi-layer feature aggregation, and [11] proposed a densely connected TDNN to increase information flow through dense connections. Temporal aggregation and sequence modeling have also been refined. Multi-level pooling was presented in [5] to combine information from different network depths. Self-attentive pooling was explored in [7], and self multi-head attention for speaker recognition was proposed in [8], allowing the model to focus on multiple salient temporal patterns. To better capture long-range dependencies, TDNN-LSTM hybrids were studied in [6], building on the LSTM formulation in [9]. Loss functions strongly influence the geometry of the embedding space. ArcFace, or Additive Angular Margin Softmax, was proposed in [4] to impose an angular margin between classes and has been successfully transferred from face to speaker recognition. Building on previous work that combines convolutional encoders with multi-head attention and metric learning [12], the present study adopts a hybrid TDNN-BiLSTM encoder with multi-head attention pooling and AAM-Softmax to obtain more discriminative speaker embeddings.

III. PROPOSED METHOD

The proposed system builds on the traditional TDNN framework while introducing two key enhancements: the integration of Bidirectional LSTM (BiLSTM) layers and Multi-Head Attention (MHA) pooling, combined with an Additive Angular Margin Softmax (AAM-Softmax / ArcFace) optimization objective. In smart environments, speaker verification operates within a broader pipeline rather than as an isolated model. Audio from a session, captured by a device under a given context, is transformed by the hybrid TDNN-BiLSTM model into an embedding, and a context-aware decision module converts similarity scores into final accept/reject outcomes.

A. TDNN-BiLSTM Backbone

Our framework extends the x-vector paradigm by integrating Time-Delay Neural Network (TDNN) layers with Bidirectional Long Short-Term Memory (BiLSTM) units, creating a hybrid encoder that captures both local and global

temporal patterns in speech. By looking at a small number of past and future frames, TDNNs effectively capture short-term dependencies such as phoneme transitions and formant variations. While TDNNs capture local information, they do not retain longer temporal dependencies across entire utterances. To overcome this limitation, we incorporate a BiLSTM module after the TDNN layers. Unlike standard LSTMs, which process input in one direction, BiLSTMs scan the speech sequence in both forward and backward directions. This bidirectional modeling enables the network to capture global sequential dependencies, such as prosody, rhythm, and intonation. By allowing the TDNN to first extract clean and localized acoustic features, the BiLSTM operates on higher-level, noise-reduced representations.

A TDNN [2] applies a 1D convolution across a temporal context window:

$$h_t = \sigma(W \cdot [x_{t-c}, \dots, x_t, \dots, x_{t+c}] + b) \quad (1)$$

where x_t is the input frame at time t , c is the context size, W and b are learnable parameters, and $\sigma(\cdot)$ is a non-linear activation (ReLU). This operation captures short-term temporal dependencies such as phoneme transitions and spectral dynamics.

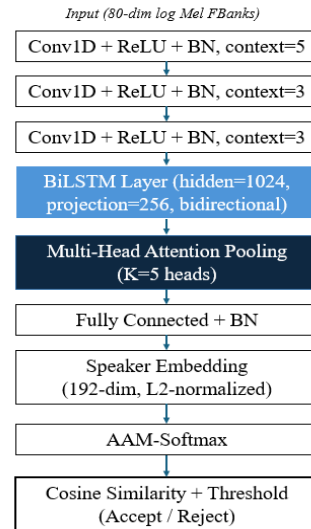


Fig. 1. Network topology of the proposed TDNN-BiLSTM with Multi-Head Attention (MHA) Pooling.

The BiLSTM [13] captures long-range dependencies by processing the sequence in both directions:

$$h_t^{BiLSTM} = [\vec{h}_t; \overleftarrow{h}_t] \quad (2)$$

where, $\vec{h}_t$: forward hidden state (past $\rightarrow$ future), $\overleftarrow{h}_t$: backward hidden state (future $\rightarrow$ past) and concatenation combines both directions.

B. Multi-Head Attention Pooling

Statistics pooling computes simple averages such as mean and variance, treating all frames equally and ignoring the fact that some frames may carry more speaker-discriminative

information than others. In contrast, the proposed system introduces a Multi-Head Attention (MHA) pooling layer [12]. This mechanism computes several independent attention distributions over the sequence of [8] frame-level features, with each head focusing on a different subset of acoustic cues.

$$z_k = \sum_{t=1}^T \alpha_t^{(k)} h_t W_k \quad (4)$$

Where, $h_t \in R^C$ is the frame-level feature at time t , $\alpha_t^{(k)}$ are the attention weights for head k , W_k : projects the values for head k . The final pooled representation, typically 192 dimensions in our framework, serves as the fixed-length speaker embedding.

C. AAM-Softmax Objective

To obtain more discriminative embeddings for open-set verification, we employ the Additive Angular Margin Softmax (ArcFace) loss [4]. Let x be the L2-normalized speaker embedding and W_j be the L2-normalized weight vector for class j . The cosine of the angle between x and W_j is $\cos \theta_j = W_j^T x$. For the ground-truth class y , AAM-Softmax adds a fixed angular margin m and uses a scaling factor s :

$$L_{AAM} = -\log \frac{e^{s \cos(\theta_y + m)}}{e^{s \cos(\theta_y + m)} + \sum_{j \neq y} e^{s \cos \theta_j}} \quad (4)$$

where θ_y is the angle between the embedding x and the weight vector W_y of the ground-truth class, m is the additive angular margin, and s is a scaling factor that controls the concentration of the softmax distribution.

D. Training and Evaluation Pipeline

Input audio is first converted into 80-dimensional log-Mel filterbank features. To improve robustness, data augmentation is applied using the MUSAN corpus (noise, music, and speech) and the RIRS_NOISES dataset (room impulse responses) to simulate noisy and reverberant conditions. The augmented features are then passed through the stacked TDNN layers and the BiLSTM block, which jointly capture local temporal patterns and long-range sequential dependencies. Multi-Head Attention pooling is subsequently used to aggregate the frame-level outputs into a fixed-length utterance representation. A fully connected layer with batch normalization projects this representation into a 192-dimensional L2-normalized speaker embedding. This embedding is trained using the AAM-Softmax loss over the set of training speakers, enforcing angular margins between classes in the embedding space. Speaker verification is performed using cosine similarity scoring between enrollment and test embeddings on the standard VoxCeleb1 test trials. A decision threshold on the similarity score is applied to produce the final Accept/Reject outcome.

IV. EXPERIMENTAL SETUP

A. Dataset

The proposed framework was trained on the VoxCeleb1 development set [10], which contains over one hundred thousand utterances from 1,211 speakers collected under

diverse acoustic and environmental conditions. This large-scale corpus provides sufficient speaker variability for learning robust speaker embeddings. Evaluation was performed on the VoxCeleb1 test set, using the standard 37,720 verification trials involving 40 speakers

B. Feature Extraction

Input audio was processed into 80-dimensional log Mel filterbank (FBank) features using a 25 ms analysis window with a 10 ms frame shift. Each training segment was two seconds long, corresponding to approximately 200 frames, obtained either by random cropping or padding as needed.

C. Data Augmentation

To simulate realistic acoustic conditions, data augmentation was employed during training. Additive noise and background speech were sampled from the MUSAN corpus, which includes noise, speech, and music recordings. In addition, reverberation effects were generated using the RIRS_NOISES dataset, which provides simulated room impulse responses to model a variety of reverberant environments.

D. Training Configuration

The model was trained using the Adam optimizer with an initial learning rate of 0.001, decayed by a factor of 0.97 every evaluation step. The batch size was set to 400 utterances per iteration to ensure a diverse mix of speakers within each batch. The speaker embedding dimension was fixed at 192, and training was conducted for a total of 80 epochs using the AAM-Softmax loss over the training speakers.

E. Evaluation Protocol

For verification, speaker embeddings were extracted for each test utterance, and similarity was measured using cosine distance. Performance was reported in terms of Equal Error Rate (EER), which reflects the operating point where false acceptance and false rejection rates are equal, and minimum Detection Cost Function (minDCF), which provides a weighted measure of verification performance under application-specific operating points.

V. RESULTS

The results demonstrate that the proposed hybrid model consistently outperforms the baselines. The integration of BiLSTM layers into the TDNN backbone enhances the modeling of long-range sequential dependencies, which complements the local feature extraction of TDNN. Furthermore, replacing attentive statistics pooling with Multi-Head Attention (MHA) pooling enables the network to capture richer and more diverse temporal representations. In addition, adopting the AAM-Softmax objective further improves the discriminative power of the speaker embeddings by enforcing an angular margin between speaker classes.

TABLE I. EER AND minDCF PERFORMANCE COMPARISON

Model	Feature	Pooling	Loss	EER (%)	minDCF
GMM-UBM [1]	MFCC	-	-	15.0	-
CNN + MHA [14]	MFCC	MHA	Triplet	14.73	-
CNN + MHA [8]	MFCC	MHA	Softmax	4.00	-
x-vector (baseline) [2]	FBank	Stats	Softmax	4.16	0.420
Proposed (ours)	FBank	MHA	AAM-Softmax	0.97	0.0717

TABLE II. ABLATION STUDY PERFORMANCE COMPARISON

Pooling	Loss	EER (%)	minDCF
Attn Stats	Softmax	1.05	0.076
Attn Stats	Triplet	1.02	0.075
MHA	Softmax	1.00	0.073
MHA	AAM-Softmax	0.97	0.0717

Finally, the combination of BiLSTM-based long-range modeling, MHA pooling, and the AAM-Softmax objective yields the best overall performance, achieving an EER of 0.97% and a minDCF of 0.0717.

VI. CONCLUSION

This paper presented a hybrid speaker verification framework that enhances TDNN-based embeddings with Bidirectional LSTM layers, Multi-Head Attention pooling, and an Additive Angular Margin Softmax (AAM-Softmax) optimization objective. By addressing the limitations of conventional pooling strategies and standard classification-based losses, the proposed system effectively captures both local and long-range temporal dependencies while enforcing angular margins in the embedding space. Experimental evaluation on the VoxCeleb1 test set demonstrates that the hybrid model achieves an Equal Error Rate (EER) of 0.97% and a minDCF of 0.0717, outperforming strong TDNN-based baselines such as the x-vector architecture.

Despite these gains, some limitations remain. The current study focuses exclusively on acoustic embeddings and does

not explicitly model channel mismatch or cross-lingual variability. Future work will investigate multimodal extensions that incorporate visual information (e.g., lip movements and facial cues) to improve robustness under noisy or cross-channel conditions.

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